

Generative AI for Patient-Centric Healthcare

Practical Tools for Doctors and Pharma Professionals

A 150-Minute Masterclass Development Program — Theory (60 min) + Hands-On Lab (90 min)

MDP WORKSHOP KIT

CLINICIAN-IN-THE-LOOP

EVIDENCE-BASED

Generative AI is transforming healthcare — from patient communication to clinical documentation, from pharmacovigilance to medical research. This program equips doctors, hospital leaders, pharma/medical affairs professionals, healthcare educators, and clinical researchers with practical, ethically grounded tools to harness AI responsibly. Every session emphasizes the **clinician-in-the-loop** principle: AI supports, but never replaces, professional judgment.

This workshop draws on guidance from the **WHO AI in Health (2023)**, **FDA AI/ML-Enabled Medical Device Action Plan**, **NEJM AI**, **PubMed**, and peer-reviewed literature. All tools and datasets in this program use synthetic or de-identified data only. No identifiable patient information should ever be entered into public AI tools.

Why GenAI Matters in Healthcare and Pharma

Healthcare generates an estimated **2,314 exabytes** of data annually. Clinicians spend up to **2 hours on documentation for every 1 hour of direct patient care** (Annals of Internal Medicine, 2016). GenAI offers a path to reclaim time, reduce cognitive load, and improve patient communication — but only when deployed with clear guardrails and realistic expectations.



The Scale of the Challenge

Physicians face exponential growth in medical literature (2.5M+ PubMed articles/year), complex regulatory demands, and rising patient expectations. Burnout affects over 50% of U.S. physicians (Mayo Clinic, 2023).



What GenAI Can Do

Summarize clinical notes, draft patient education materials in multiple languages, assist literature reviews, generate differential diagnosis checklists, triage adverse event reports, and automate routine documentation — all at clinician direction.



What GenAI Cannot Do

Replace clinical judgment, make autonomous treatment decisions, guarantee accuracy without verification, or handle identifiable patient data in public tools. AI hallucinations remain a documented risk (NEJM AI, 2024).



Core Principle: GenAI is a clinician-in-the-loop assistant. Every AI output requires professional verification before clinical or regulatory use. This principle is non-negotiable throughout this program.

Tool Landscape: 12 Categories of GenAI for Healthcare

The following matrix categorizes the GenAI tools covered in this program across 12 functional domains. Tools are marked as **Free**, **Freemium**, or **Paid**, with accessibility notes for India/global use. All clinical tools require clinician verification before use.

Category	Key Tools	Tier	India Access	Best Demo Use	Risk Level	Privacy Warning
A. Patient Communication	ChatGPT, Gemini, Claude	Free/Freemium	Yes (VPN may be needed)	Discharge note simplification	Medium	⚠️ No PHI
B. Multilingual Education	Gemini, ChatGPT, DeepL	Free/Freemium	Yes	Hindi/Tamil counseling script	Low	✅ Safe (synthetic)
C. Clinical Documentation	Copilot, ChatGPT, Doximity	Freemium/Paid	Varies	SOAP note generation	High	⚠️ No PHI
D. Clinical Reasoning	OpenEvidence, Glass Health	Freemium	Limited	DDx checklist with citations	High	⚠️ Verify all outputs
E. Research & Literature	Elicit, Consensus, Semantic Scholar, Connected Papers, ResearchRabbit	Free/Freemium	Yes	PICO + evidence table	Low	✅ Public data only
F. Pharma Medical Affairs	Claude, ChatGPT, Paperpal	Freemium	Yes	MI response drafting	High	⚠️ Regulatory review required
G. Pharmacovigilance	ChatGPT, Julius AI	Free/Freemium	Yes	AE case triage (synthetic)	High	⚠️ Synthetic data only
H. Clinical Trials	ChatGPT, Elicit	Free/Freemium	Yes	Eligibility criteria summary	Medium	⚠️ No patient identifiers
I. Data Analysis	Julius AI, ChatGPT Code Interpreter	Freemium	Yes	Patient feedback dashboard	Medium	⚠️ De-identified only
J. Presentation & Design	Gamma, Canva, Napkin AI	Free/Freemium	Yes	Slide deck from outline	Low	✅ Safe
K. Hospital Operations	Copilot, ChatGPT	Freemium	Varies	Workflow automation draft	Medium	⚠️ No operational PHI
L. Ethics & Governance	All tools + policy frameworks	N/A	Global	Bias audit exercise	High	⚠️ Institutional policy required

Sources: WHO Guidance on AI in Health (2023); FDA AI/ML Action Plan (2021–2023); NEJM AI (2024); PubMed; Official tool documentation. All tools listed are current as of Q1 2025. Accessibility in India may require VPN for some U.S.-only services.

Patient-Centric Applications of GenAI

GenAI's most transformative potential lies in improving how clinicians communicate with and educate patients. These applications reduce health literacy barriers, support shared decision-making, and enable culturally sensitive counseling — all while keeping the clinician in control of the final message.



Discharge Summary Simplification

Convert complex clinical discharge summaries into plain-language, patient-friendly instructions at a 6th-grade reading level. AI can generate multilingual versions and include visual medication schedules. **Tool:** ChatGPT or Claude with a structured prompt. **Risk:** Always verify dosing instructions against the original prescription.



Multilingual Patient Counseling

Generate counseling scripts in Hindi, Tamil, Spanish, or 50+ languages for chronic disease management (diabetes, hypertension, COPD). AI can adapt tone for cultural context and literacy level. **Tool:** Gemini or ChatGPT. **Risk:** Verify medical accuracy in translation — automated translation may miss clinical nuance.

Shared Decision-Making Aids

Create balanced, evidence-based patient information sheets comparing treatment options (e.g., surgery vs. conservative management). AI can summarize pros, cons, and probabilities from current guidelines. **Tool:** OpenEvidence or Consensus for evidence-backed outputs. **Risk:** Ensure all statistics are current and cited.



Patient Journey Mapping

Use AI to draft personalized patient journey maps for chronic conditions — outlining key milestones, warning signs, follow-up schedules, and self-management goals. **Tool:** ChatGPT or Gamma for visual outputs. **Risk:** Journey maps must be reviewed by the treating clinician before sharing.

Facilitator Question: Ask participants: "Have you ever had a patient who didn't understand their discharge instructions? How did you find out, and what was the consequence?" Use responses to anchor the value of AI-assisted communication.

Doctor Productivity: Clinical Documentation & Reasoning

Clinical documentation consumes a disproportionate share of physician time. GenAI can draft, summarize, and structure notes — but every output requires clinician review. This section covers the most high-impact documentation use cases with safety guardrails built in.

Documentation Workflows

OPD Note Summarization

Input: Raw dictated or typed notes. Output: Structured, concise OPD summary with chief complaint, HPI, assessment, and plan. **Tool:** ChatGPT, Copilot. **Warning:** Never input identifiable patient data into public AI tools — use synthetic or fully de-identified cases only.

SOAP Note Generation

Provide clinical findings; AI structures into Subjective, Objective, Assessment, Plan format. Saves 5–10 minutes per patient encounter. **Tool:** Doximity Ask, ChatGPT. **Warning:** SOAP notes are legal documents — physician must review and sign off before entry into the EHR.

Referral Letter Drafting

AI drafts referral letters with relevant history, current medications, and specific clinical questions for the specialist. **Tool:** Claude, ChatGPT. **Warning:** Verify all medication lists and dates against the patient record.

Prior Authorization Draft

Generate insurance prior authorization letters citing clinical guidelines and medical necessity criteria. **Tool:** ChatGPT with guideline prompts. **Warning:** Payer-specific criteria vary — always confirm against current payer policies.

Clinical Reasoning Support

Differential Diagnosis Checklist

Input: Patient presentation (de-identified). Output: Ranked differential with supporting and refuting features, red flags, and suggested workup. **Tool:** OpenEvidence, Glass Health. **Warning:** AI-generated DDX is a starting point, not a conclusion. Clinical correlation is mandatory.

Red-Flag Extraction

AI scans clinical notes to flag concerning features: weight loss + anemia, new headache in elderly, chest pain characteristics. **Tool:** ChatGPT with structured prompt. **Warning:** False negatives possible — AI is not a substitute for clinical vigilance.

Follow-Up Plan Generation

AI drafts structured follow-up instructions: timeline, monitoring parameters, warning signs, and when to return. **Tool:** ChatGPT, Gemini. **Warning:** Follow-up intervals must align with clinical guidelines and individual patient risk.



Clinician Verification Required: All AI-generated clinical content must be reviewed, edited, and approved by a licensed physician before use. AI outputs may contain hallucinations, outdated information, or inappropriate recommendations.

Pharma Applications: Medical Affairs, Safety & Trials

Pharma and medical affairs professionals face unique regulatory constraints when using GenAI. All outputs must pass through medical, legal, and regulatory (MLR) review. This section focuses on AI-assisted drafting and analysis — not autonomous decision-making.



Medical Information Responses

AI drafts evidence-based responses to unsolicited medical information queries. Outputs include citations from PubMed and are flagged for MLR review. **Tools:** Claude, ChatGPT with citation prompts. **Compliance:** All responses require Medical Affairs sign-off before external use. Never use AI for on-label promotional content.



Adverse Event Triage

AI categorizes synthetic AE reports by seriousness, expectedness, and causality using MedDRA terminology. Speeds initial case processing for PV teams. **Tools:** ChatGPT, Julius AI. **Compliance:** Use synthetic data only in public tools. Real AE data requires validated, compliant PV systems (e.g., Argus, ArisG).



KOL Briefing Documents

AI synthesizes published literature on a therapeutic area to create concise KOL briefing packs: key opinion leaders' publications, research interests, and potential collaboration areas. **Tools:** Elicit, Semantic Scholar, ResearchRabbit. **Compliance:** Verify all publication data and ensure no confidential information is included.



Clinical Trial Eligibility Summaries

AI extracts and simplifies inclusion/exclusion criteria from protocols into plain-language summaries for site teams and potential investigators. **Tools:** ChatGPT, Elicit. **Compliance:** Protocol language is legally binding — AI summaries are for internal use only, not for patient-facing materials without regulatory review.



Regulatory-Safe Communication

AI assists in drafting internal scientific communications, slide decks, and briefing documents. All outputs are flagged for regulatory review before external distribution. **Tools:** Paperpal, Grammarly, ChatGPT. **Compliance:** Pharma communications are subject to FDA/EMA promotional regulations — AI does not replace regulatory review.

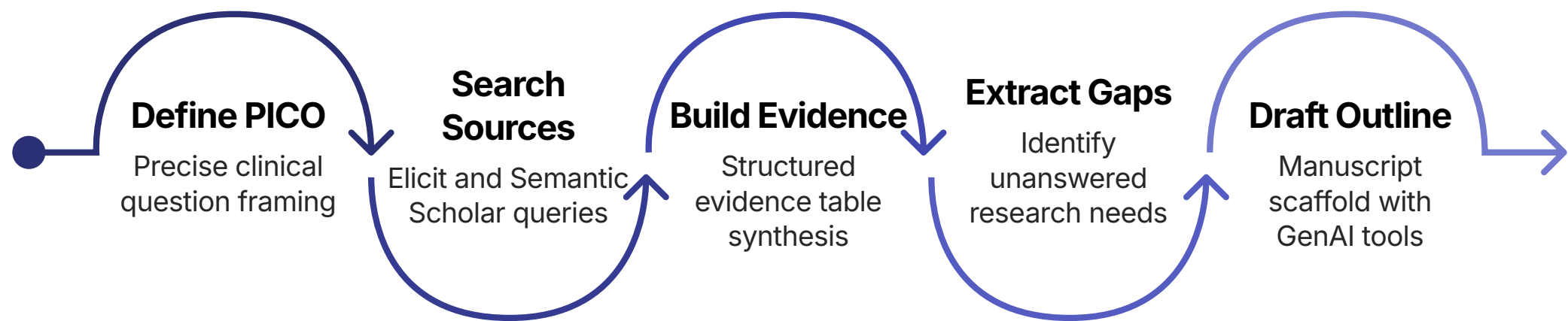


Literature Summary for MSL Teams

AI summarizes new publications in a therapeutic area, extracting key findings, limitations, and clinical implications for MSL field teams. **Tools:** Consensus, Elicit, SciSpace. **Compliance:** Summaries must be verified against original publications. MSLs are responsible for accurate scientific exchange.

Research Applications: Literature Review to Manuscript

GenAI is reshaping how researchers conduct literature reviews, formulate research questions, and draft manuscripts. These tools dramatically reduce time-to-insight — but researchers must maintain rigorous standards for citation accuracy, plagiarism avoidance, and methodological transparency.



This workflow illustrates how AI tools can accelerate each stage of the research process — from initial question formulation through manuscript preparation — while maintaining researcher oversight at every step.

AI Research Tools by Function

→ PICO Question Generation

Input your clinical question; AI structures it into Population, Intervention, Comparison, Outcome format optimized for database searches. **Tools:** ChatGPT, Elicit. **Tip:** Always refine the AI-generated PICO against your actual research scope.

→ Automated Literature Search

Elicit, Consensus, and Semantic Scholar use AI to search, rank, and summarize papers by relevance — not just keyword matching. Connected Papers and ResearchRabbit visualize citation networks. **Best for:** Rapid scoping reviews and systematic review planning.

→ Evidence Table Generation

AI extracts study design, sample size, outcomes, and limitations from abstracts into structured comparison tables. **Tools:** Elicit, SciSpace. **Warning:** Always verify extracted data against the full text — AI may misinterpret statistical results.

→ Manuscript Writing Support

Paperpal, Jenni AI, and Grammarly assist with academic writing, grammar, and journal-specific formatting. **Warning:** AI should not write your manuscript — it should assist your writing. Many journals now require AI disclosure statements (COPE, 2023).

Reference Management & Organization

Zotero (free, open-source) integrates with browser plugins to capture citations from PubMed, Google Scholar, and journal websites. AI plugins can auto-tag and categorize references by theme. **Best practice:** Use Zotero groups for collaborative projects; enable automatic backup to avoid data loss.

Research Gap Extraction

AI analyzes a set of abstracts to identify what questions remain unanswered — a critical step for grant applications and study justification. **Tool:** ChatGPT with a structured prompt: "Based on these 20 abstracts, what are the top 5 unanswered research questions?"

AI Disclosure & Ethics

COPE (Committee on Publication Ethics) and major journals (Nature, Science, NEJM) now require explicit disclosure of AI use in manuscript preparation. AI cannot be listed as an author. Researchers must verify all AI-generated citations — hallucinated references are a documented risk (Nature, 2023).

☐ **Key Rule:** AI assists research — it does not conduct it. The researcher is responsible for every claim, citation, and conclusion in a published manuscript.

Safety, Privacy, Ethics & Governance

Responsible AI use in healthcare requires understanding the regulatory landscape, privacy risks, bias concerns, and institutional governance requirements. These are not optional considerations — they are foundational to ethical practice.

Regulatory Frameworks

WHO AI in Health Guidance (2023)

Six guiding principles: protect human autonomy, promote well-being, ensure transparency, foster responsibility, ensure inclusiveness, and promote sustainable AI. Applies to all AI developers and deployers in health.

FDA AI/ML Action Plan

FDA regulates AI/ML-enabled medical devices through the 510(k), De Novo, and PMA pathways. The FDA's "Good Machine Learning Practice" (GMLP) provides 10 guiding principles for AI developers.

HIPAA & Data Privacy

Public AI tools (ChatGPT, Gemini, Claude) are **not** HIPAA-compliant. Never enter Protected Health Information (PHI) into these platforms. Use only synthetic or fully de-identified data for training and demos.

Key Risk Categories

● Hallucination Risk

AI may generate plausible but incorrect clinical information, fabricated citations, or inaccurate drug dosing. **Mitigation:** Always verify outputs against authoritative sources (UpToDate, PubMed, drug labels).

● Bias & Equity

AI models trained on non-representative data may produce biased recommendations for underrepresented populations. **Mitigation:** Audit AI outputs for demographic bias; use diverse training data where possible.

● Privacy Breach

Entering patient data into public AI tools may constitute a reportable data breach. **Mitigation:** Institutional AI policies must explicitly prohibit PHI input into non-compliant tools.

● Over-Reliance

Clinicians may defer to AI outputs without adequate scrutiny. **Mitigation:** Training must emphasize the clinician-in-the-loop principle and build verification habits.

⚠ **Discussion Cases for Group Debate:** (1) A patient tells their doctor they asked ChatGPT about stopping their medication — how do you respond? (2) A hospital wants to deploy an AI chatbot for appointment triage — what are the risks? (3) A researcher uses AI to write a literature review — is this acceptable? Use these cases to explore the boundary between AI assistance and AI replacement.

90-Minute Hands-On Lab: 6 Exercises

Participants work through six structured exercises using synthetic datasets and the GenAI tools covered in the theory session. Each exercise includes a facilitator guide, expected outputs, and debrief questions. All datasets are synthetic — no real patient data is used.

01

Exercise 1: Discharge Note Simplification (15 min)

Dataset: `discharge_summaries.csv` (synthetic). **Task:** Use ChatGPT or Claude to rewrite a complex discharge summary into plain language at a 6th-grade reading level, including a medication schedule and warning signs. **Prompt:** "Rewrite this discharge summary for a patient with 8th-grade literacy. Include: what happened, medications with timing, warning signs, follow-up appointment." **Debrief:** Compare outputs across participants — what worked? What required manual correction?

03

Exercise 3: Differential Diagnosis with Safety Checks (15 min)

Dataset: Synthetic chest pain case (de-identified). **Task:** Use OpenEvidence or Glass Health to generate a differential diagnosis checklist. Then use ChatGPT with a chain-of-verification prompt: "List the differential, then for each diagnosis list one finding that supports it AND one finding that argues against it." **Debrief:** Compare AI output with participant clinical judgment — where did AI miss something important?

05

Exercise 5: AI-Assisted Literature Review (15 min)

Topic: "GLP-1 agonists for weight loss in non-diabetic patients." **Task:** Use Elicit to find and summarize 5 relevant papers. Use Connected Papers to visualize the citation network. Build a mini evidence table. **Tool:** Elicit + Connected Papers + Zotero. **Debrief:** Did AI miss any landmark papers? How does AI-assisted review compare to manual PubMed search in speed and quality?

02

Exercise 2: Multilingual Counseling Script (15 min)

Dataset: `outpatient_cases.csv` (synthetic diabetes case). **Task:** Generate a counseling script in Hindi or Spanish for a newly diagnosed Type 2 diabetes patient covering diet, medications, monitoring, and when to seek help. **Tool:** Gemini or ChatGPT. **Debrief:** Discuss translation accuracy, cultural appropriateness, and whether a native speaker should review before clinical use.

04

Exercise 4: Adverse Event Triage (15 min)

Dataset: `adverse_event_reports.csv` (30 synthetic AE cases). **Task:** Use ChatGPT or Julius AI to categorize each AE by seriousness (serious/non-serious), expectedness (expected/unexpected), and causality (related/not related). **Tool:** Julius AI for batch analysis. **Debrief:** Discuss limitations of AI in PV — why human review is irreplaceable for regulatory submissions.

06

Exercise 6: Patient Feedback Analysis & Dashboard (15 min)

Dataset: `patient_feedback.csv` (50 synthetic satisfaction surveys). **Task:** Upload to Julius AI or ChatGPT Code Interpreter. Ask: "Analyze sentiment by department, identify top 3 complaints, and create a summary dashboard." **Tool:** Julius AI or ChatGPT Advanced Data Analysis. **Debrief:** Discuss how AI-driven feedback analysis can inform quality improvement initiatives — and what human context is still needed.

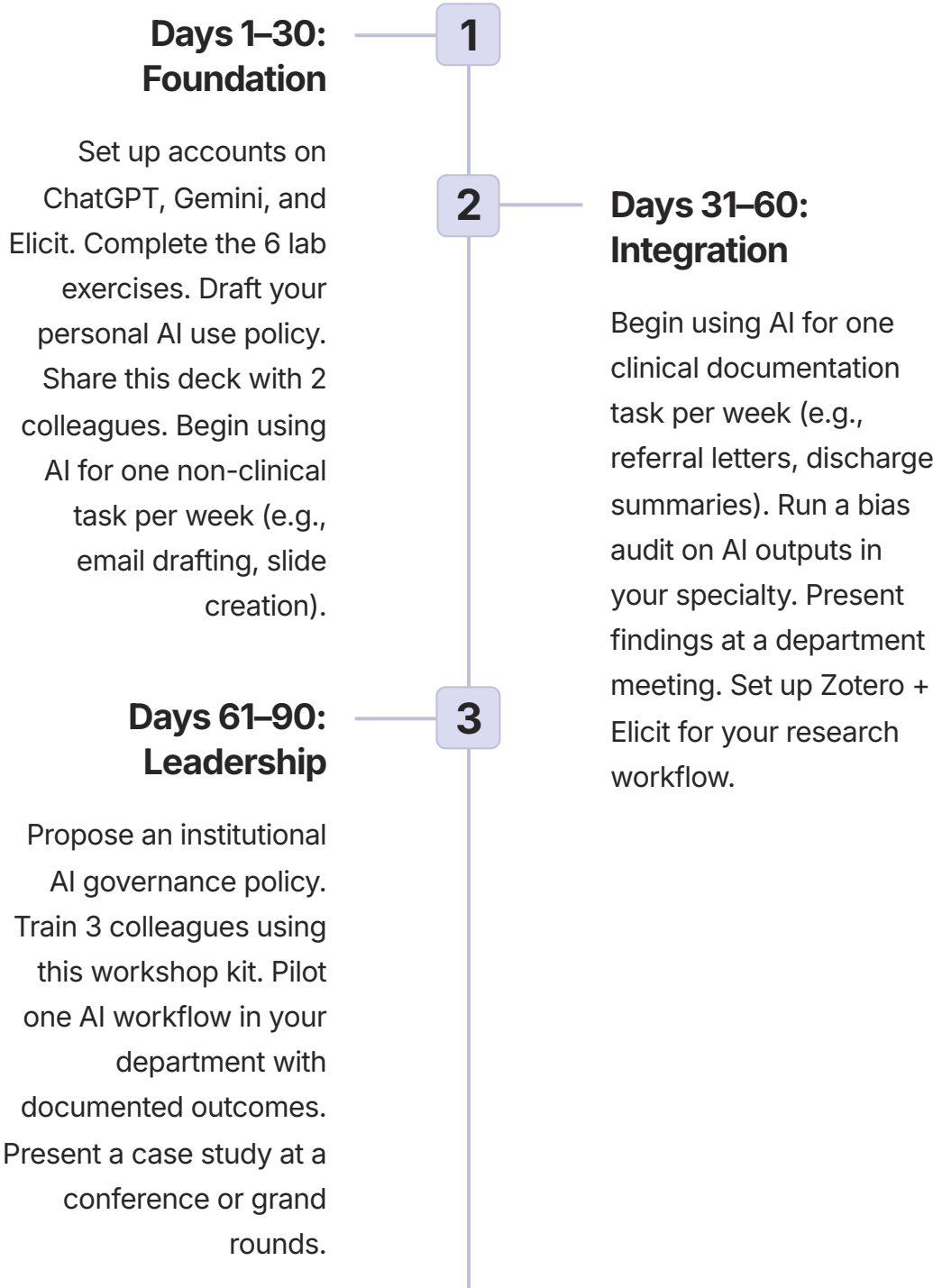
Prompt Bank, Take-Home Roadmap & Implementation Checklist

This final section provides a curated prompt bank organized by skill level and use case, plus a 30-60-90 day implementation roadmap for participants to begin integrating GenAI into their practice responsibly.

Essential Prompt Patterns

1	2	3	4
Beginner Prompt Use Case: Patient education material. Prompt: "Explain [condition] to a patient in simple language. Cover: what it is, causes, symptoms, treatment options, and when to see a doctor." Common Mistake: Too vague — specify literacy level and language. Improved: "Explain hypertension to a 60-year-old patient with 8th-grade literacy in English. Use short sentences. Include 3 lifestyle changes."	Chain-of-Verification Prompt Use Case: Clinical reasoning. Prompt: "Generate a differential diagnosis for [presentation]. For each diagnosis, list: (1) one finding that supports it, (2) one finding that argues against it, (3) one test to confirm or rule out." Why it works: Forces AI to self-audit rather than presenting a one-sided list.	Safety-Aware Clinical Prompt Use Case: Any clinical task. Prompt: "This is a synthetic/de-identified case for educational purposes. [Case details]. Provide [output type]. Flag any assumptions you are making and note where clinical judgment is required." Key: Explicitly stating the data is synthetic reduces the risk of accidental PHI input.	Pharma-Compliant Prompt Use Case: Medical information response. Prompt: "Draft a balanced, evidence-based response to an unsolicited medical query about [drug]. Include: mechanism, approved indications, key safety data from [source], and a disclaimer that this requires Medical Affairs review before use. Cite PubMed IDs where possible."

30-60-90 Day Implementation Roadmap



Final Implementation Checklist

- ✓

Tool Setup Complete

ChatGPT, Gemini, Elicit, Zotero, Julius AI accounts created and tested with synthetic data.
- ✓

Safety Rules Internalized

I will never enter PHI into public AI tools. I will verify all clinical AI outputs before use. I understand AI hallucination risks.
- ✓

First Use Case Identified

I have identified one specific task I will automate with AI this week, with a written prompt ready.
- ✓

Governance Conversation Started

I have shared this resource with my team and initiated a conversation about institutional AI policy.
- ✓

Workshop Complete! You now have the tools, prompts, datasets, and framework to begin using GenAI responsibly in your practice. Remember: AI is your assistant — you are the clinician. Stay curious, stay critical, and stay patient-centered.